

# Problem Set 4: Regression Discontinuity Designs

MY457 | WT 2026 | Week 9

Add your answers to this .qmd file by replacing the dedicated placeholder text. When you are finished, render the .qmd file as a .pdf and make sure you push your work to GitHub.

## 1 Concepts

- 1.1 Consider a data generating process with  $i \in [1, \dots, N]$  units. Treatment  $D_i$  is assigned based on values of a variable  $X_i$ , such that  $D = \mathbf{1}\{X_i > c\}$ , where  $c$  is a cutpoint. What is the precise name of the research design we could use here? Briefly explain your answer.

### Answer

We can analyse this data using a Sharp Regression Discontinuity Design (SRDD). It's sharp (not fuzzy) because the assignment to treatment is a deterministic function of the running variable  $X_i$  and the cutpoint  $c$ .

- 1.2 Below is an incomplete table of potential outcomes for the data generating process described above. Fill in the missing values and **bold** the values you fill in. Assume a cutpoint of  $c = 1$ .

| $i$ | $X_i$ | $D_i$ | $Y_{0i}$ | $Y_{1i}$ | $Y_i$ | $\tau_i$ |
|-----|-------|-------|----------|----------|-------|----------|
| 1   | 0.5   | ?     | 10       | 15       |       |          |
| 2   | 1.5   | ?     | 20       | 30       |       |          |
| 3   | 0.8   | ?     | 15       | 20       |       |          |
| 4   | 1.2   | ?     | 25       | 35       |       |          |

💡 Answer

| $i$ | $X_i$ | $c$ | $D_i$    | $Y_{0i}$ | $Y_{1i}$ | $Y_i$     | $\tau$    |
|-----|-------|-----|----------|----------|----------|-----------|-----------|
| 1   | 0.5   | 1   | <b>0</b> | 10       | 15       | <b>10</b> | <b>5</b>  |
| 2   | 1.5   | 1   | <b>1</b> | 20       | 30       | <b>30</b> | <b>10</b> |
| 3   | 0.8   | 1   | <b>0</b> | 15       | 20       | <b>15</b> | <b>5</b>  |
| 4   | 1.2   | 1   | <b>1</b> | 25       | 35       | <b>35</b> | <b>10</b> |

1.3 Explain the difference between the local randomisation perspective and the continuity perspective in regression discontinuity designs.

💡 Answer

**Local Randomisation:** Within a small window around the cutpoint  $c$ , units are as-if randomly assigned a value of  $X_i$ , and thus  $D_i$ .

**Continuity:** Potential outcomes are continuous at the threshold  $c$ , meaning that  $Y_{0i}$  and  $Y_{1i}$  do not “jump” at  $c$ .

1.4 Explain the two equations below. How do they correspond to the two perspectives of RDD?

1.

$$(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i \quad \text{for all } i \text{ where } |X_i - c| \leq h$$

2.

$$\lim_{\epsilon \uparrow 0} E[Y_{0i} \mid X_i = c + \epsilon] = E[Y_{0i} \mid X_i = c]$$

$$\lim_{\epsilon \downarrow 0} E[Y_{1i} \mid X_i = c + \epsilon] = E[Y_{1i} \mid X_i = c]$$

💡 Answer

**Equation 1** corresponds to the local randomisation perspective. It states that within a small window around the cutpoint ( $|X_i - c| \leq h$ ), the potential outcomes are independent of treatment assignment.

**Equation 2** corresponds to the continuity perspective. It states that the expected potential outcomes arbitrarily close to above and below the cutpoint are the same as potential outcomes directly at the cutpoint.

1.5 Which estimand can we identify with a sharp RDD? Write down the estimand in terms of potential outcomes and explain in words what it means.

 Answer

In a sharp RDD, we can identify the Local Average Treatment Effect (LATE) at the cutpoint  $c$ , which is given by:

$$\tau_{LATE} = E[Y_{1i} - Y_{0i} \mid X_i = c]$$

This is the average treatment effect for units right at the cutpoint. A key limitation is that this estimand may not be generalisable to units far from the cutpoint, as the treatment effect may vary with  $X_i$ .

1.6 Which estimands can we identify with a fuzzy RDD? Write down the estimands in terms of potential outcomes and explain in words what it means.

 Answer

In a fuzzy RDD, we can identify two estimands: the Local Average Treatment Effect for compliers (LATE) and the Local Intent-to-Treat Effect (LITT).

The **LATE for compliers** is given by:

$$\tau_{FRD} = E[Y_{1i} - Y_{0i} \mid i \text{ is a complier and } X_i = c]$$

The **LITT** is given by:

$$\tau_{LITT} = E[Y_{1i} - Y_{0i} \mid X_i = c]$$

Conceptually, the LATE is the average treatment effect for compliers at the cutpoint, while the LITT is the effect of being assigned to the treatment for units at the cutpoint.

1.7 Beyond SUTVA, Which assumptions do we need to make to identify the LATE in a fuzzy RDD?

 Answer

To identify the LATE in a fuzzy RDD, we need to make the following assumptions:

1. **Augmented Continuity:** The expected potential outcomes for the treatment ( $D_{zi}$ ) and the expected potential outcomes for the dependent variable ( $Y_{zi}$ ) are continuous at the cutpoint  $c$ .

2. **First Stage/Relevance:** The probability of receiving treatment changes discontinuously at the cutpoint, meaning that there is a jump in the probability of treatment at  $c$ .
3. **Monotonicity:** There are no defiers, meaning that there are no units that would take the treatment if assigned to control and vice versa.
4. **Exclusion Restriction:** The cutpoint only affects the outcome through its effect on treatment assignment, meaning that there are no other channels through which the cutpoint can affect the outcome.

1.8 Suppose a city summons drivers caught exceeding 50 mph at automated speed cameras to appear in traffic court. However, not all summoned drivers actually appear — some ignore the summons, and some drivers below the threshold are summoned for other reasons. A researcher wants to know the causal effect of a court appearance on subsequent traffic compliance, measured as the number of traffic violations by that driver in the following two years. Using data from drivers with speeds within a narrow bandwidth around the cutoff, the researcher estimates two regressions with the recentred running variable  $\tilde{X}_i = X_i - c$  (where  $c = 50$ ) and an indicator  $Z_i = \mathbf{1}(X_i > 50)$ .  $Y_i = \alpha + \tau Z_i + \beta \tilde{X}_i + \epsilon_i$ . The estimated coefficient  $\tau$  is -0.4. They also estimate a second regression with the same recentred variable and indicator, but with an indicator for whether the driver appeared in court as the dependent variable,  $D_i$ . The estimated coefficient on  $Z_i$  in this second regression is 0.6. What is the LITT and what is the LATE? What do the findings tell us about the research question?

#### Answer

The LITT is simply the coefficient  $\hat{\tau}$  from the first regression, which is -0.4. This means that being assigned to appear in court (i.e., being caught speeding) on average reduces the number of traffic violations by drivers at the cutpoint (going 50mph) by 0.4 over the next two years.

The LATE can be calculated using the ratio of the two coefficients from the regressions. The first regression gives us the effect of being assigned to treatment (court appearance) on the outcome (traffic violations), while the second regression gives us the effect of being assigned to treatment on the probability of actually receiving treatment (appearing in court). The LATE is calculated as:

$$\tau_{LATE} = \frac{\tau_{LITT}}{\tau_{LITT_D}} = \frac{-0.4}{0.6} \approx -0.67$$

This means that for compliers at the threshold (drivers who would appear in court if they were recorded at speeds over 50 mph and would not appear if they weren't), the average treatment effect of appearing in court is a reduction of approximately 0.67 traffic

violations in the following two years.

1.9 Evaluate the plausibility of the above design.

#### Answer

One concern could be that speeders who don't appear in court might receive a harsher (or additional) penalty than those who comply with the summons. For example, they might receive a higher fine or have their licence suspended whereas those who appear in court might have the opportunity to contest the ticket or receive a lesser penalty. The effect of  $Z$  on  $Y$  would then not only be through the effect of court appearance. Another concern might be that drivers could be able to sort below the threshold, some drivers are very skilled at edging the speed limits.

## 2 Simulations

2.1 Explain the simulation code below.

```
n <- 1000

set.seed(123)
X <- rnorm(n, mean = 0, sd = 15)
c <- 0

Y0 <- 100 + 0.45 * X - 0.15 * (X^2) + 0.001 * (X^3) + rnorm(n, mean = 0, sd = 25)
Y1 <- 175 + 0.25 * X + 0.15 * (X^2) - 0.001 * (X^3) + rnorm(n, mean = 0, sd = 25)

D <- ifelse(X > c, 1, 0)
Y <- ifelse(D == 1, Y1, Y0)

dat <- tibble(
  X,
  D,
  Y,
  Y0,
  Y1
)
```

### Answer

This code simulates a dataset for a sharp regression discontinuity design. It generates 1000 observations of a running variable  $X$  from a normal distribution with mean 0 and standard deviation 15. The cutpoint  $c$  is set to 0. The potential outcomes  $Y_0$  and  $Y_1$  are generated as nonlinear functions of  $X$  with some added random noise. The treatment assignment  $D$  is determined by whether  $X$  exceeds the cutpoint  $c$ . The observed outcome  $Y$  is then assigned based on the treatment status.

2.2 Calculate the true ATE and the true LATE. How do these two quantities compare?

### Answer

The true ATE can be calculated as the average difference between the potential outcomes across all units:

```
mean(dat$Y1 - dat$Y0)
```

```
[1] 138.9717
```

Calculating the true LATE is a bit more complex because we don't have observations *exactly* at the cutpoint. However, we can use our knowledge of the data generating process to calculate the true LATE by plugging in the cutpoint value (0) into the equations generating the potential outcomes. Because all but the first term in the equations for  $Y_0$  and  $Y_1$  are functions of  $X$  and  $X = 0$  at the cutpoint, all that's left is:

```
E_Y0_at_c <- 100
E_Y1_at_c <- 175
LATE <- E_Y1_at_c - E_Y0_at_c
LATE
```

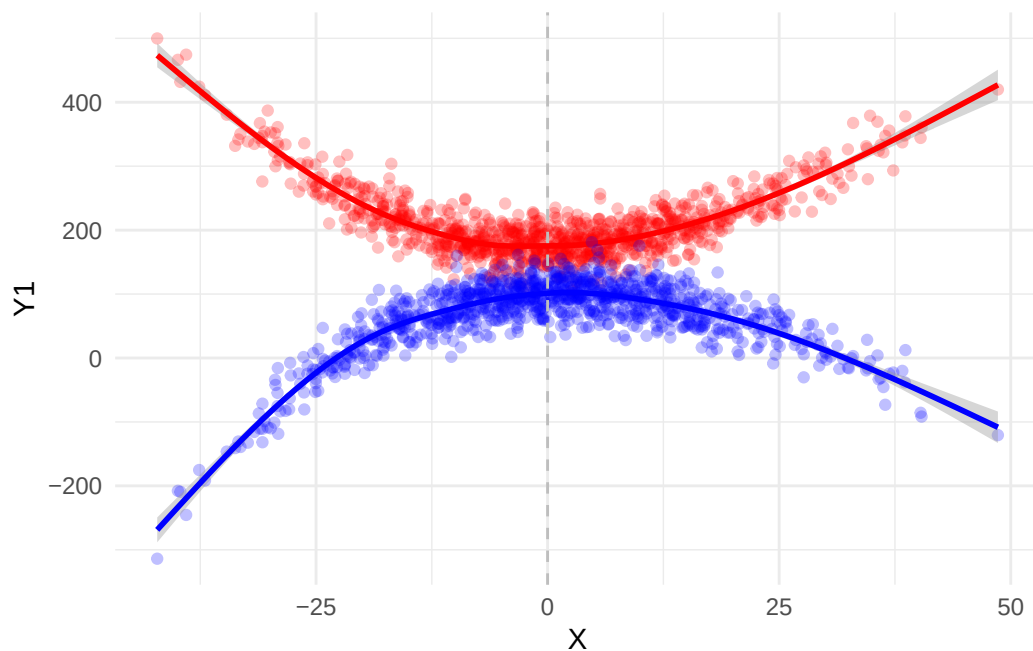
```
[1] 75
```

Our LATE is 75 whereas the ATE is  $\approx 139$ . The difference between the two quantities implies that the treatment effect varies with the running variable  $X$ . If we were to extrapolate the LATE to the entire population, we would be severely underestimating the true ATE.

2.3 In a clear, well formatted, and compelling figure, visualise *both* potential outcomes and how they relate to the cutpoint.

💡 Answer

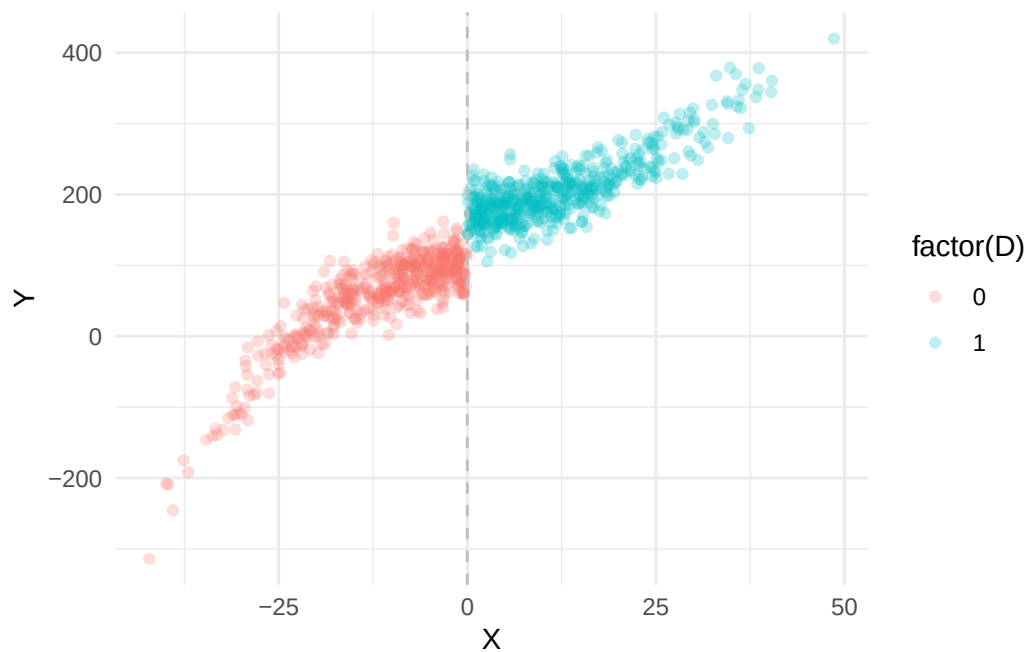
```
ggplot(dat) +  
  aes(x = X, y = Y1) +  
  geom_point(color = "red", alpha = 0.25) +  
  geom_smooth(color = "red") +  
  geom_point(aes(x = X, y = Y0), color = "blue", alpha = 0.25) +  
  geom_smooth(aes(x = X, y = Y0), color = "blue") +  
  geom_vline(xintercept = c, linetype = "dashed", color = "gray") +  
  theme_minimal()
```



2.4 In a clear, well formatted, and compelling figure, visualise only the observed Y.

💡 Tip

```
ggplot(dat) +  
  aes(x = X, y = Y, color = factor(D)) +  
  geom_point(alpha = 0.25) +  
  geom_vline(xintercept = c, linetype = "dashed", color = "gray") +  
  theme_minimal()
```



2.5 Use a global (do not focus on a narrow window) linear regression with common slopes to estimate the LATE. What do you find? How does this estimator perform?

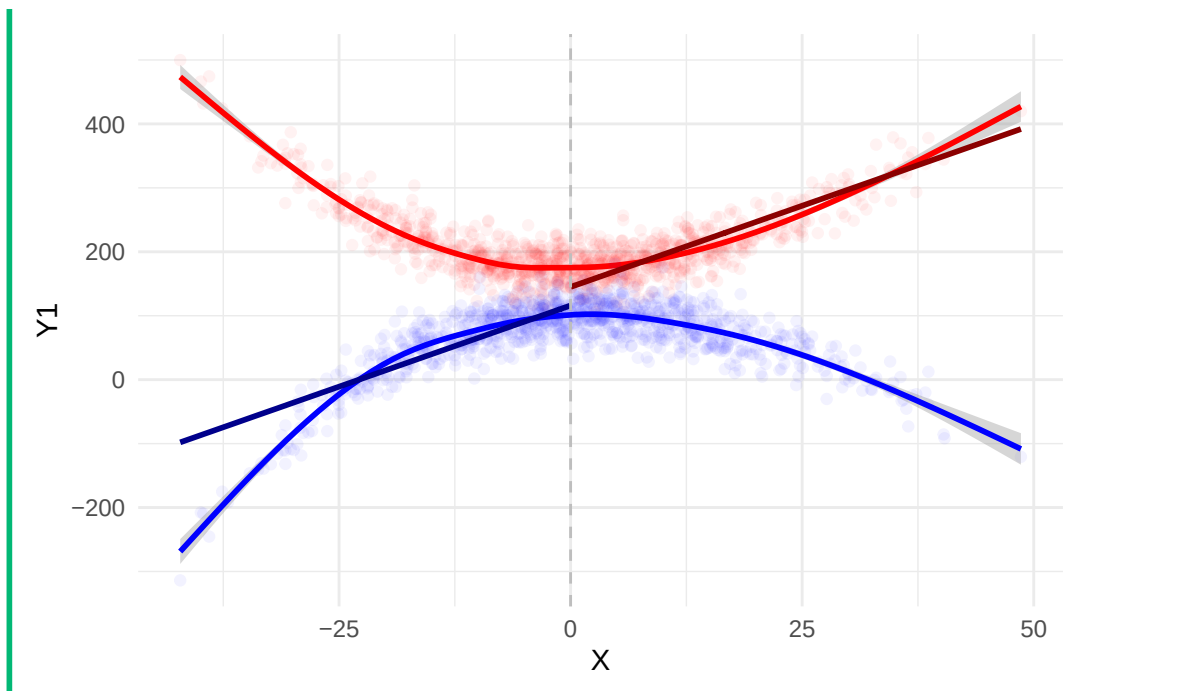
💡 Answer

```
lm(Y ~ X + D, data = dat) |>  
  modelsummary(  
    output = "markdown",  
    stars = T,  
    gof_map = c("nobs", "r.squared")  
  )
```

|             | (1)                   |
|-------------|-----------------------|
| (Intercept) | 116.045***<br>(1.872) |
| X           | 5.080***<br>(0.108)   |
| D           | 28.967***<br>(3.203)  |
| Num.Obs.    | 1000                  |
| R2          | 0.889                 |

We estimate a LATE of 28.9, which is a severe underestimate of the true LATE (75). This estimator performs poorly because it imposes a linear functional form on the relationship between the running variable and the outcome, which is not correct in this case. We can visualise this by plotting the fitted values from the regression against the true potential outcomes:

```
dat <- dat |>
  mutate(fitted = predict(lm(Y ~ X + D, data = dat)))
ggplot(dat) +
  aes(x = X, y = Y1) +
  geom_point(color = "red", alpha = 0.05) +
  geom_smooth(color = "red") +
  aes(x = X, y = Y0, color = "blue", alpha = 0.05) +
  geom_smooth(aes(x = X, y = Y0), color = "blue") +
  geom_line(
    data = dat[dat$X < c, ], aes(x = X, y = fitted),
    color = "darkblue", size = 1
  ) +
  geom_line(
    data = dat[dat$X >= c, ], aes(x = X, y = fitted),
    color = "darkred", size = 1
  ) +
  geom_vline(xintercept = c, linetype = "dashed", color = "gray") +
  theme_minimal()
```



2.6 Use a global (do not focus on a narrow window) polynomial regression with varying slopes to estimate the LATE. Pick an adequate polynomial order based on your knowledge about the DGP. What do you find? How does the estimator perform? How might you expect this estimator to perform outside of a simulated setting?

## 💡 Answer

```
modp2 <- lm(
  Y ~ X * D + D * I(X^2),
  data = dat
)
modp3 <- lm(
  Y ~ X * D + D * I(X^2) + D * I(X^3),
  data = dat
)
modp4 <- lm(
  Y ~ X * D + D * I(X^2) + D * I(X^3)
    + D * I(X^4),
  data = dat
)
modp7 <- lm(
  Y ~ X * D + D * I(X^2) + D * I(X^3)
    + D * I(X^4) + D * I(X^5) + D * I(X^6)
    + D * I(X^7),
  data = dat
)

modelsummary(
  list(
    "2nd order" = modp2,
    "3rd order" = modp3,
    "4th order" = modp4,
    "7th order" = modp7
  ),
  output = "markdown",
  stars = T,
  coef_map = c("D" = "LATE"),
  gof_map = c("nobs", "r.squared")
)
```

|          | 2nd order            | 3rd order            | 4th order            | 7th order            |
|----------|----------------------|----------------------|----------------------|----------------------|
| LATE     | 76.739***<br>(3.491) | 76.021***<br>(4.430) | 73.998***<br>(5.439) | 82.823***<br>(8.931) |
| Num.Obs. | 1000                 | 1000                 | 1000                 | 1000                 |
| R2       | 0.931                | 0.931                | 0.931                | 0.931                |

As you can see, if we pick a polynomial close to the one used in the true DGP (2nd, 3rd, or 4th order) our estimate is quite close to the true LATE (75). However, if we pick a polynomial that is too flexible (7th order), we get a very poor estimate of the LATE. This is due to overfitting – the estimator captures noise rather than the true underlying signal. Hence, the most flexible model doesn't necessarily perform better.

2.7 Let's now analyse our data with the `rdrobust` package. Estimate the LATE with local polynomial approximation. You may leave the settings at their defaults, or change settings if you so choose. What do you find?

#### 💡 Answer

```
# This is just a helper function to print the results of rdrobust in a nice format.
print_rdrobust_table <- function(model) {
  cbind(
    as.data.frame(model$coef),
    as.data.frame(model$se),
    as.data.frame(model$ci)
  ) |>
  mutate(
    across(c(`CI Lower`, `CI Upper`), \(x) round(x, 2)),
    CI = paste0("[", `CI Lower`, ", ", `CI Upper`, "]")
  ) |>
  select(Coeff, `Std. Err.`, CI) |>
  kableExtra::kable(
    col.names = c("Coefficient", "Standard Error", "95% CI"),
    digits = 3,
    format.args = list(big.mark = ",")
  )
}

# Estimate the model
rdrobust::rdrobust(y = Y, x = X, c = c) |>
  print_rdrobust_table()
```

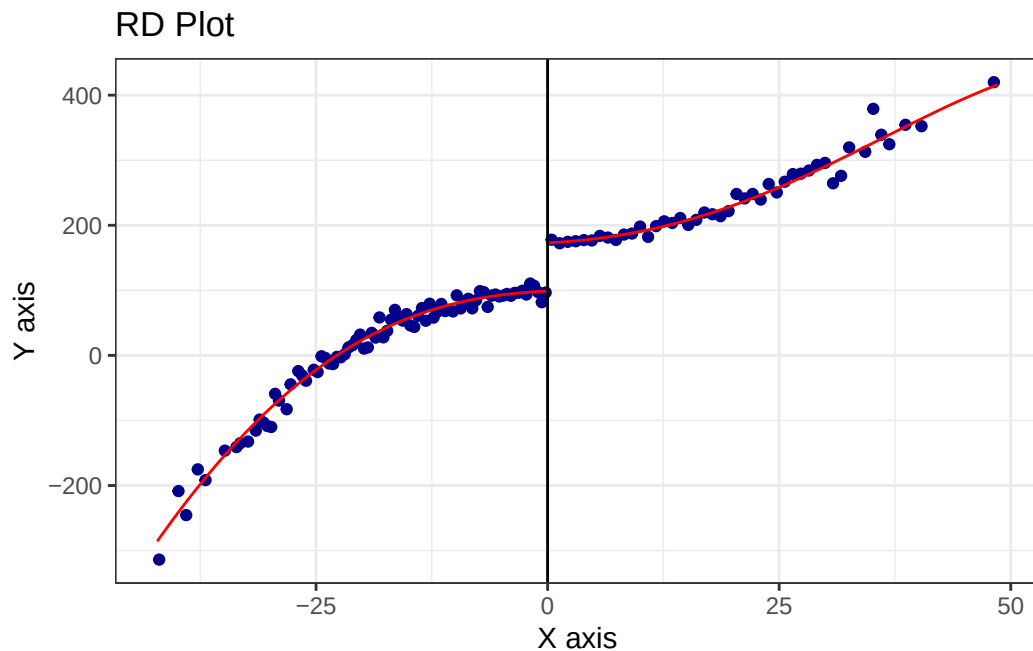
|                | Coefficient | Standard Error | 95% CI         |
|----------------|-------------|----------------|----------------|
| Conventional   | 74.838      | 4.976          | [65.09, 84.59] |
| Bias-Corrected | 77.113      | 4.976          | [67.36, 86.87] |
| Robust         | 77.113      | 5.492          | [66.35, 87.88] |

The conventional estimator does a really good job at estimating the LATE and yields 74.8. The bias corrected estimator is a little bit further from the true ATE but still reasonably close and with confidence intervals that include the true LATE.

2.8 Use the `rdplot` function within the package to create a regression discontinuity plot.

💡 Answer

```
rdrobust::rdplot(y = Y, x = X, c = c)
```



### 3 Replication

In this section, we will use real-world data to reinforce what we have learned. We will analyse the dataset employed in *Rural Roads and Local Economic Development*. Physical connectivity through paved roads may be a catapult for economic development. The authors of this paper study the economic impact of a \$40 billion national programme designed to build roads across rural India.

- 3.1 Skim the paper and fill out the table below to provide a summary of the paper's research design. Be as precise and concise as possible. Edit the table directly in your .qmd to provide your answers.

| Category                | Description |
|-------------------------|-------------|
| Research Question       |             |
| Unit of Analysis        |             |
| Population              |             |
| Treatment               |             |
| Identification Strategy |             |
| Running Variable        |             |
| cutpoint(s)             |             |
| Estimand                |             |

#### Answer

| Category                | Description                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research Question       | How does rural road access affect local economic development?                                                                                                           |
| Unit of Analysis        | Rural villages                                                                                                                                                          |
| Population of Interest  | Rural villages in India that didn't have paved road in 2001, are located in states complying with PMGSY guidelines and have populations near the 500 or 1,000 cutpoints |
| Treatment               | Receipt of a new paved all-weather road under the PMGSY program                                                                                                         |
| Identification Strategy | Fuzzy Regression Discontinuity                                                                                                                                          |
| Running Variable        | Population of the village                                                                                                                                               |
| cutpoint(s)             | 500 and 1000                                                                                                                                                            |
| Estimand                | LATE (ATE for compliers at the threshold)                                                                                                                               |

- 3.2 Import the dataset `data/pmgys_working_aer_mainsample_reduced.dta` and assign it to an object called `df`. The table below contains the variable names and their descriptions. Use this information to understand the dataset and answer the questions below.

| Variable Name  | Description                                                                         |
|----------------|-------------------------------------------------------------------------------------|
| v_pop          | Population in 2001 normalised to the nearest cutpoint                               |
| r2012          | Indicator for existence of a rural road in 2012                                     |
| *_index_andrsn | Indices for local economic outcomes, the outcomes reported in Table 3 of the paper. |

#### 💡 Answer

```
df <- read_dta("data/pmgsy_working_aer_mainsample_reduced.dta")
```

3.3 Test the first stage with a plot and a simple regression of the treatment on the running variable and the indicator for being above the cutpoint. What do you find?

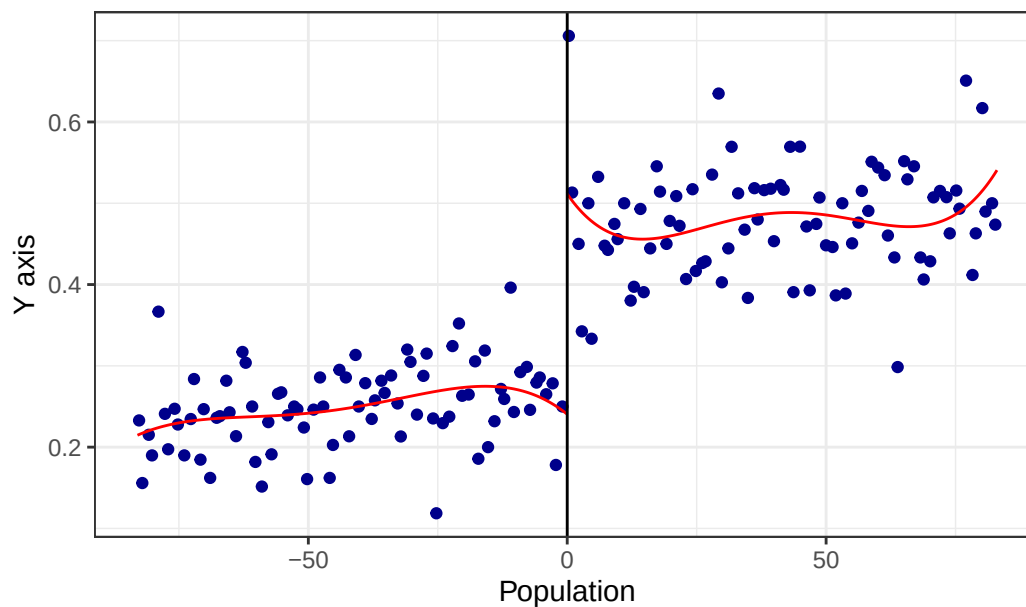
#### 💡 Answer

Let's first visualise the relationship between the running variable and the treatment assignment using `rdplot`:

```
rdrobust::rdplot(y = df$r2012, x = df$v_pop, x.label = "Population")
```

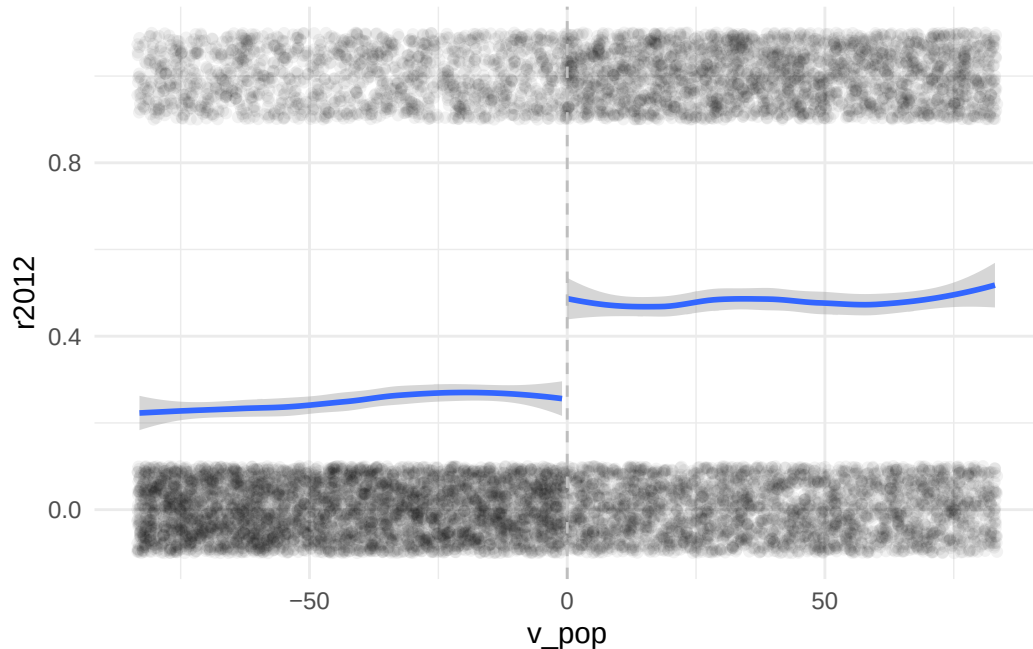
```
[1] "Mass points detected in the running variable."
```

RD Plot



We could create an even simpler plot by just plotting the raw data:

```
ggplot(df) +  
  aes(x = v_pop, y = r2012) +  
  geom_point(alpha = 0.05, position = position_jitter(height = .1)) +  
  geom_smooth(data = df[df$v_pop < 0, ], method = "loess") +  
  geom_smooth(data = df[df$v_pop >= 0, ], method = "loess") +  
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray") +  
  theme_minimal()
```



As you can see, there is a clear jump in the probability of receiving the treatment at the cutpoint, which suggests that the first stage is strong. Units below the threshold have a treatment probability of  $\approx 0.22$ , units above the threshold have a treatment probability of  $\approx 0.45$ . We can also test whether the treatment probability is the same left and right of the cutpoint with a simple linear regression:

```
lm(r2012 ~ v_pop * I(v_pop >= 0), data = df) |>  
  modelsummary(  
    output = "markdown",  
    stars = T,  
    gof_map = c("nobs", "r.squared")  
  )
```

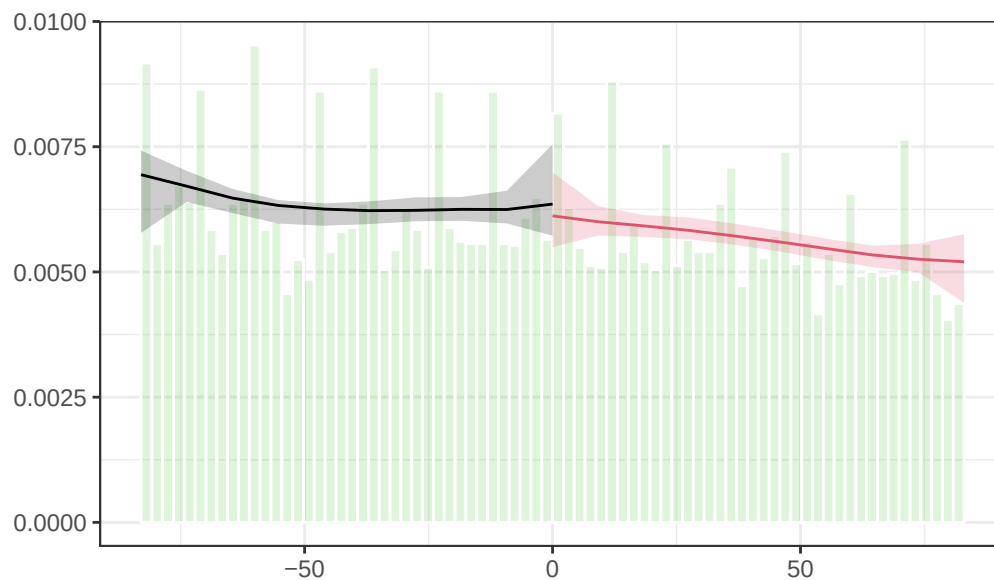
|                           | (1)                 |
|---------------------------|---------------------|
| (Intercept)               | 0.276***<br>(0.012) |
| v_pop                     | 0.001*<br>(0.000)   |
| I(v_pop >= 0)TRUE         | 0.191***<br>(0.017) |
| v_pop × I(v_pop >= 0)TRUE | -0.000<br>(0.000)   |
| Num.Obs.                  | 11432               |
| R2                        | 0.058               |

Again, we find a significant and large (compare with the coef. of v\_pop) coefficient for the indicator of being above the cutpoint.

3.4 A common concern in RD designs using population thresholds is manipulation at the threshold (see [Eggers et al. \(2017\)](#)). Using the `rddensity::rdplotdensity()` function, test for sorting in around the cutpoint. What do you find?

💡 Answer

```
rddensity::rdplotdensity(rddensity::rddensity(df$v_pop), df$v_pop, noPlot = T)$Estplot
```



The plot shows no discontinuity in the density of villages around the cutpoint. We find no evidence of sorting.

- 3.5 Estimate the LITT for the transport index with `rdrobust`. Let the package choose the optimal bandwidth and the optimal polynomial order. Interpret the results and the parameters chosen by the package (you may ignore all parameters related to bias estimation).

#### 💡 Answer

```
litt_transport <- rdrobust::rdrobust(y = df$transport_index_andrsn, x = df$v_pop)
litt_transport |> summary()
```

Sharp RD estimates using local polynomial regression.

|                     |            |        |
|---------------------|------------|--------|
| Number of Obs.      | 11432      |        |
| BW type             | mserd      |        |
| Kernel              | Triangular |        |
| VCE method          | NN         |        |
| Number of Obs.      | 6018       | 5414   |
| Eff. Number of Obs. | 2232       | 2169   |
| Order est. (p)      | 1          | 1      |
| Order bias (q)      | 2          | 2      |
| BW est. (h)         | 31.050     | 31.050 |
| BW bias (b)         | 48.393     | 48.393 |
| rho (h/b)           | 0.642      | 0.642  |
| Unique Obs.         | 83         | 84     |

|           | Point<br>Estimate | Robust<br>z | Inference<br>P> z | [ 95% C.I. ]     |
|-----------|-------------------|-------------|-------------------|------------------|
| RD Effect | 0.063             | 0.606       | 0.545             | [-0.111 , 0.209] |

The LITT is estimated to be 0.063, which means that villages right above the cutpoint score 0.063 units higher on the transportation index than units just below the cutpoint. However, with a standard error of 0.069 and p-value of 0.36, we fail to reject the null hypothesis of no effect at conventional significance levels. The optimal bandwidth  $h$

chosen by the package is 31 and the optimal polynomial order  $p$  is 1. By default, a triangular kernel was used.

- 3.6 Now estimate the LATE for the transport index with `rdrobust`. Use a bandwidth of 84 and leave the other parameters to their default values. Interpret the result and compare it to the one's reported in the paper.

#### 💡 Answer

```
rdrobust::rdrobust(y = df$transport_index_andrsn, x = df$v_pop, fuzzy = df$year2012, h = 84)
summary()
```

Fuzzy RD estimates using local polynomial regression.

|                     |            |        |
|---------------------|------------|--------|
| Number of Obs.      | 11432      |        |
| BW type             | Manual     |        |
| Kernel              | Triangular |        |
| VCE method          | NN         |        |
| Number of Obs.      | 6018       | 5414   |
| Eff. Number of Obs. | 6018       | 5414   |
| Order est. (p)      | 1          | 1      |
| Order bias (q)      | 2          | 2      |
| BW est. (h)         | 84.000     | 84.000 |
| BW bias (b)         | 84.000     | 84.000 |
| rho (h/b)           | 1.000      | 1.000  |
| Unique Obs.         | 6018       | 5414   |

First-stage estimates.

|           | Point<br>Estimate | Robust Inference<br>z | P> z  | [ 95% C.I. ]    |
|-----------|-------------------|-----------------------|-------|-----------------|
| Rd Effect | 0.198             | 7.840                 | 0.000 | [0.161 , 0.268] |

Treatment effect estimates.

|           | Point<br>Estimate | Robust<br>z | Inference<br>P> z | [ 95% C.I. ]     |
|-----------|-------------------|-------------|-------------------|------------------|
| RD Effect | 0.474             | 1.241       | 0.215             | [-0.227 , 1.009] |

Using a fuzzy RDD with a bandwidth of 84, we estimate a LATE of 0.474. This means that for compliers at the threshold (villages arbitrarily close to the cutpoint that would receive a new road if assigned to treatment and would not receive a new road if assigned to control), the average treatment effect of receiving a paved road is an increase of 0.474 units on the transportation index. With a standard error of 0.221 and a p-value of 0.3, this estimate is significant at  $\alpha = 0.05$ . The paper reports a LATE of 0.41 with a standard error of 0.187 and a p-value of 0.03. The difference between our estimate and the one reported is small and could be due to the conditioning set used by the authors, who control for “baseline village-level controls for amenities and economic indicators, as well as district-cutpoint fixed effects”.

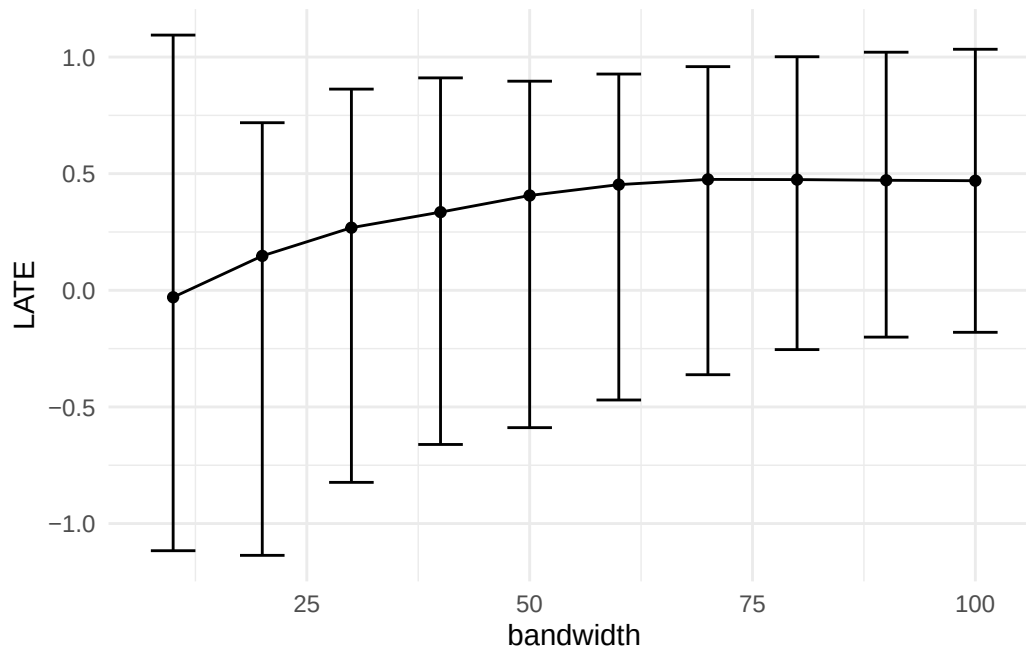
- 3.7 The authors used a bandwidth of 84 for their main analysis. The `rdrobust` suggested an optimal bandwidth of 31. Write a simple loop that estimates the LATE for the transport index with `rdrobust` for a range of bandwidths `seq(10, 100, by = 10)`. Plot the results with the bandwidth on the x-axis and the estimated LATE on the y-axis. What do you find?

## 💡 Answer

```
bandwidths <- seq(10, 100, by = 10)
results <- data.frame(
  bandwidth = bandwidths,
  LATE = NA,
  se = NA,
  p_value = NA,
  ci_lower = NA,
  ci_upper = NA
)

for (i in seq_along(bandwidths)) {
  model <- rdrobust::rdrobust(
    y = df$transport_index_andrsn,
    x = df$v_pop,
    fuzzy = df$r2012,
    h = bandwidths[i]
  )
  results$LATE[i] <- model$coef[1]
  results$se[i] <- model$se[1]
  results$p_value[i] <- model$pv[1]
  results$ci_lower[i] <- model$ci[3, 1]
  results$ci_upper[i] <- model$ci[3, 2]
}

ggplot(results) +
  aes(x = bandwidth, y = LATE) +
  geom_point() +
  geom_line() +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), width = 5) +
  theme_minimal()
```



As we increase the bandwidth, we consider more and more observations further away from the cutpoint. As we increase the bandwidth, we include more observations further from the cutoff. This reduces variance (visible in the shrinking confidence intervals) but introduces bias. In our plot, we see that smaller bandwidths yield smaller LATEs with larger confidence intervals, while bandwidths between 70 and 100 yield larger LATEs with smaller confidence intervals. Only for the estimates with a bandwidth of 70 and above, the confidence intervals do not include zero.

- 3.8 Use the `RDHonest::RDHonest()` function to estimate the LATE for the transport index. Leave all arguments at their package defaults. Interpret the results and compare them to the ones obtained with `rdrobust`.

## 💡 Answer

```
rd_honest_default <- RDHonest::RDHonest(  
  formula = transport_index_andrsn | r2012 ~ v_pop,  
  data = df  
)  
  
rd_honest_default$coefficients |>  
  t() |>  
  as.data.frame() |>  
  rownames_to_column("Parameter") |>  
  kableExtra::kable(  
    col.names = c("Parameter", "Estimate"),  
    digits = 3,  
    format.args = list(big.mark = ",")  
  )
```

| Parameter          | Estimate           |
|--------------------|--------------------|
| term               | Fuzzy RD parameter |
| estimate           | 0.2381897          |
| std.error          | 0.3052468          |
| maximum.bias       | 0.2184008          |
| conf.low           | -0.4853267         |
| conf.high          | 0.961706           |
| conf.low.onesided  | -0.4822975         |
| conf.high.onesided | 0.9586768          |
| bandwidth          | 25.24444           |
| eff.obs            | 2956.235           |
| leverage           | 0.001944381        |
| cv                 | 2.370267           |
| alpha              | 0.05               |
| method             | Holder             |
| M                  | 0.003451194        |
| M.rf               | 0.000610593        |
| M.fs               | 0.001016306        |
| first.stage        | 0.2470642          |
| kernel             | triangular         |
| p.value            | 0.541507           |

The confidence intervals are very wide and include zero. **RDHonest** estimated a first-stage  $M$  of 0.001 and a reduced form  $M$  of 0.0006. Hence, even if the conditional expectation function is close to linear at the cutpoint and the worst case bias is small, the estimate is not statistically significant. In other words, even when bounding the bias relatively narrowly, we cannot reject the null hypothesis of no effect.

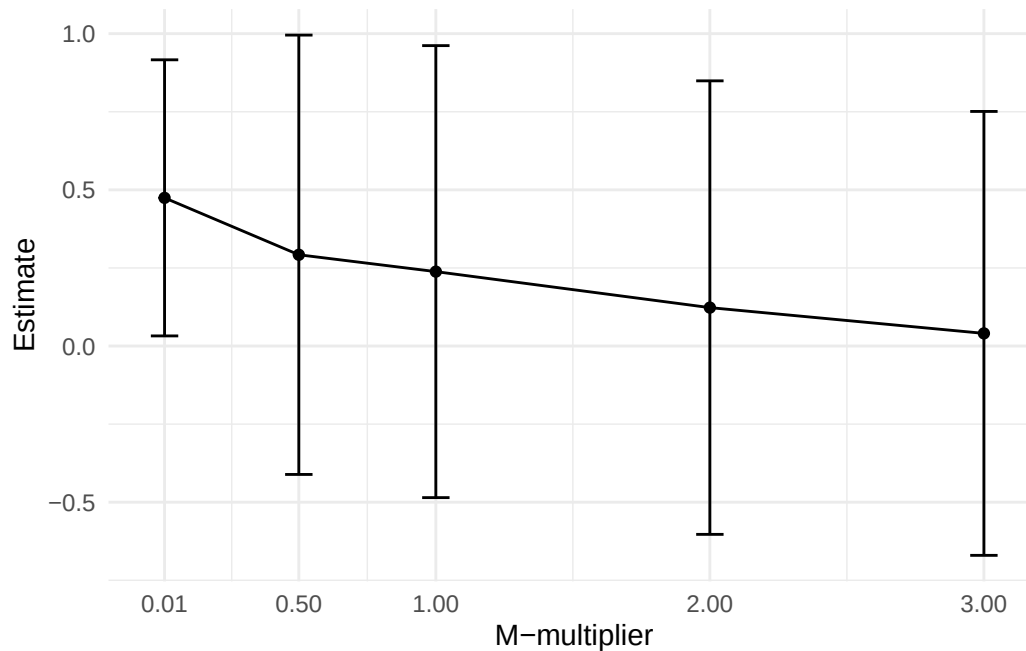
- 3.9 BONUS 1: Using the default  $M$ -values estimated by **RDHonest** in the previous question as a baseline, repeat the estimation with these values multiplied by  $c(0.01, 0.5, 1, 2, 3)$ . Create an informative plot that shows how the estimate and confidence intervals change as we change  $M$ .

## 💡 Answer

```
M_multis <- c(0.01, 0.5, 1, 2, 3)
results_honest <- tibble(
  M = M_multis,
  Estimate = NA,
  CI_Lower = NA,
  CI_Upper = NA
)

for (i in seq_along(M_multis)) {
  model <- RDHonest::RDHonest(
    formula = transport_index_andrsn | r2012 ~ v_pop,
    data = df,
    M = M_multis[i] * c(rd_honest_default$coefficients$M.rf, rd_honest_default$coefficients$M.v)
  )
  results_honest$Estimate[i] <- model$coefficients$estimate[1]
  results_honest$CI_Lower[i] <- model$coefficients$conf.low[1]
  results_honest$CI_Upper[i] <- model$coefficients$conf.high[1]
}

ggplot(results_honest) +
  aes(x = M_multis, y = Estimate) +
  geom_point() +
  geom_line() +
  geom_errorbar(aes(ymin = CI_Lower, ymax = CI_Upper), width = 0.1) +
  scale_x_continuous(
    breaks = M_multis
  ) +
  xlab("M-multiplier") +
  theme_minimal()
```



The the larger the M multiplier, the wider the confidence intervals. The CIs only exclude 0 if we set the M-multiplier to 0.01 (which means that the worst case bias is  $0.01 * 0.001 = 0.00001$  for the reduced form and  $0.01 * 0.0006 = 0.000006$  for the first stage). In essence, only if we assume that the conditional expectation function is practically linear at the cutpoint, can we reject the null hypothesis of no effect.

Note that the point estimates shift as we change the M-multiplier. This is because the bandwidth is chosen optimally given the M-multiplier and thus differs between models.

3.10 BONUS 2: Use placebo cutpoints at -50, -25, 25, and 50 to estimate the LITT for the transport index using `rdrobust()`. What do you find?

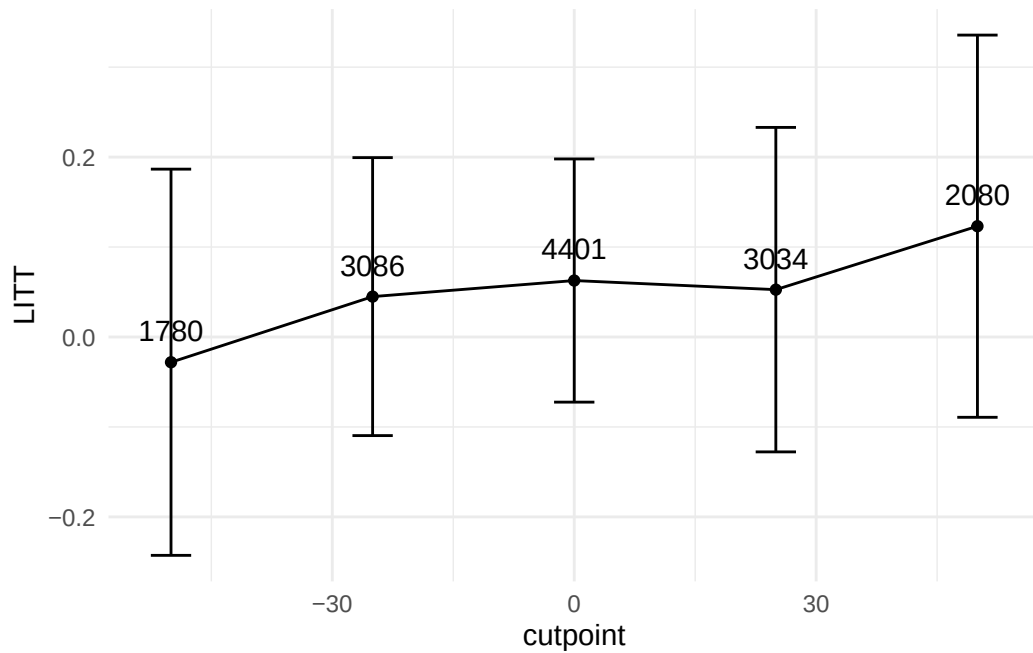
## 💡 Answer

```
placebo_cutpoints <- c(-50, -25, 0, 25, 50)
placebo_results <- data.frame(
  cutpoint = placebo_cutpoints,
  LITT = NA,
  se = NA,
  p_value = NA,
  N = NA
)

for (i in seq_along(placebo_cutpoints)) {
  model <- rdrobust::rdrobust(
    y = df$transport_index_andrsn,
    x = df$v_pop,
    c = placebo_cutpoints[i]
  )
  placebo_results$LITT[i] <- model$coef[1]
  placebo_results$se[i] <- model$se[1]
  placebo_results$p_value[i] <- 2 * (1 - pnorm(abs(model$coef[1] / model$se[1])))
  placebo_results$N[i] <- sum(model$N_h)
}
placebo_results
```

|   | cutpoint | LITT        | se         | p_value   | N    |
|---|----------|-------------|------------|-----------|------|
| 1 | -50      | -0.02808588 | 0.10953193 | 0.7976287 | 1780 |
| 2 | -25      | 0.04490910  | 0.07883352 | 0.5689015 | 3086 |
| 3 | 0        | 0.06275170  | 0.06896489 | 0.3628710 | 4401 |
| 4 | 25       | 0.05262859  | 0.09202292 | 0.5673847 | 3034 |
| 5 | 50       | 0.12316867  | 0.10839774 | 0.2558452 | 2080 |

```
ggplot(placebo_results) +
  aes(x = cutpoint, y = LITT) +
  geom_point() +
  geom_line() +
  geom_text(aes(label = N), vjust = -1) +
  geom_errorbar(aes(ymin = LITT - 1.96 * se, ymax = LITT + 1.96 * se), width = 5) +
  theme_minimal()
```



The placebo cutpoints yields LITTs similar to the one we estimated at the true cutpoint. However, the confidence intervals for the estimate at  $c = 0$  are slightly smaller than those of the other cutpoints. The overall pattern is a bit ambiguous, we see that the larger the cutpoint, the larger the LITT. Generally, we would like to see a pattern where the true estimate at the true cutpoint stands out and the other cutpoints are mostly centered around zero.